

Hyeonsu Jeong

Undergraduate Student | Department of Industrial and Systems Engineering

Gyeongsang National University

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EDUCATION

- 03/2021 – 02/2027 **B.E. in Industrial and Systems Engineering**
B.E. Double Major in Big Data Convergence
GPA(Major): 3.65(3.77) / 4.5
Gyeongsang National University, Jinju, South Korea
- 09/2024 – 07/2025 **Smart Manufacturing Big Data Micro-Degree**
Gyeongsang National University, Jinju, South Korea

RESEARCH EXPERIENCE

- 08/2026 – Present **Research Intern**
Transportation & Logistics Optimization Lab, Yonsei University, Seoul, Korea
- Participating in research on transportation and logistics optimization
- 06/2025 – 02/2026 **Undergraduate Researcher**
Digital Transformation Laboratory, Gyeongsang National University, Jinju, Korea
- Developed optimization algorithms for CCTV placement and unmanned aerial vehicle (UAV) routing for visual data acquisition

RESEARCH PROJECTS

▪ Lead Researcher

- 10/2025 – 01/2026 **Optimization of Multi-Type CCTV Camera Placement for Manufacturing Site Monitoring**
Gyeongsang National University – National University Development Program
- 06/2025 – 12/2025 **AI-based CCTV Image Analysis for Worker Safety Management on Manufacturing Sites**
Gyeongsang National University – RISE Industry-Academia Collaborative Project

▪ Research Assistant

- 06/2025 – 12/2025 **Digital Twin Platform for Smart Construction and Automation**
National Research Foundation of Korea (NRF) – STEAM Research Project (Strategic Program)

PUBLICATIONS & RESEARCH OUTPUTS

▪ **Journal Articles**

International

- [1] **Jeong, H.**, Lee, S., Kim, J., Hwang, J., and Moon, S.* (2026). “Automated optimization of visual sensor deployment for construction sites.” *Automation in Construction*, 192, 107264. DOI: 10.1016/j.autcon.2026.107264 [Q1, SCIE, 2025 IF 12.6]

Domestic

- [1] **Jeong, H.**, Koo, H., Park, S., and Moon, S.*. “Knowledge Graph-Based Legal Chain Reasoning for Industrial Safety Decision Support.” *Journal of the Korean Institute of Industrial Engineers*. [Accepted, Q1, KCI, Excellent Accredited]

▪ **Conference Proceedings**

Oral

- [1] **Jeong, H.**, Lee, S., and Moon, S.* (2025). “Optimization of CCTV Camera Placement for Construction Site Monitoring.” *Oral Presentation, 2025 Fall Conference of Korea Institute of Industrial Engineers (KIIE)*, Daejeon, South Korea.

Poster

- [1] Kim, J., Hwang, J., **Jeong, H.**, Lee, S., Lee, T., and Moon, S.* (2026). “Optimization of Resolution of 3D Digital Maps in Industrial Sites Based on Work Area Importance Derived from Terrain Information.”, Korea Institute of Industrial Engineers (KIIE) Conference, Gyeongju, South Korea.
- [2] **Jeong, H.**, Lee, S., and Moon, S.* (2025). “Development of Foundational Technology for AI-based CCTV Video Analysis for Worker Safety Management on Manufacturing Sites.” Poster Presentation, 2025 The Next AI (Gyeongnam), Changwon, South Korea.

▪ **Intellectual Property Rights**

Patent

- [1] Kim, J., Moon, S., Lee, T., Jeong, M., and **Jeong, H.** (2026) “Digital Map Resolution Optimization Technology Based on Spatial Importance in Construction Sites” (Korea, Under Review)
- [2] **Jeong, H.**, Moon, S., and Kim, J. (2026. 04. 23.) “Method for Exploration on Optimal Locations of CCTV Cameras for Construction Site Monitoring.” Application No. 10-2026-0073586. (Korea)

Software

- [1] Lee, S., Moon, S., **Jeong, H.**, and Kim, D. (2025. 9. 11.) “Program for Realistic Color Matching on 3D Point Clouds of Industrial Sites.” (Korea, Program C-2025-035569)

RESEARCH INTERESTS

- Combinatorial Optimization and Decision-Making
- Spatial Optimization, Routing, and Resource Allocation
- Optimization for Logistics and Automated Systems
- Learning-based Optimization for Complex Systems

ACTIVITIES**▪ Experience**

- 05/2026 – Present **Startup-Centered University Program: “Worker-Centric Industrial Safety Monitoring and Legal Decision Support Platform.” (Grant Number: 20410310)**
Korean Government-Funded Startup Program
- 06/2026 – 07/2026 **Purdue Academy of Global Engineering: Research Program**
Edwardson School of Industrial Engineering, Purdue University, Indiana, USA.
- 12/2025 – 01/2026 **Experiential Project-Based Learning Program for International Students: Face Recognition Attendance System**
FPT IS & FPT University, Danang, Vietnam.
- 12/2025 **2025 Gyeongnam Global Innovation Festa G-NEX – Startup Booth Operation**
Gyeongnam Global Innovation Festa
- 10/2025 **Student Startup Team U-300+ – National Finals Qualifier**
Ministry of Education, Korea
- 01/2025 **Lecture at Next-Generation Startup School**
Main Contents Co., Ltd.
- 11/2024 – 02/2025 **GNU Pioneer Leaders Startup Supporters, 1st Cohort**
GNU Startup Support Center, Gyeongsang National University

▪ Leadership

- 05/2025 - ~ **CEO, Startup “Workrithm”**
Jinju, Korea.
- 01/2026 – 12/2027 **President, Startup Club “APLUSES”**
Gyeongsang National University, Jinju, Korea.
- 03/2025 – 02/2026 **Planning Director, Startup Club “APLUSES”**
Gyeongsang National University, Jinju, Korea

TECHNICAL SKILLS**▪ Optimization Algorithms**

- Greedy Algorithm, Simulated Annealing (SA), Late Acceptance Hill Climbing (LAHC), 2-opt

▪ Programming & Data Processing

- Python
- CloudCompare (Point Cloud Data Processing)

▪ Data & Systems

- Neo4j (Graph Database), Graph Modeling
- FastAPI

▪ AI / Computer Vision

- CCTV-based monitoring system development
- Face recognition (ArcFace, RetinaFace)

AWARDS

04/2026	2026 GSAT Young Startup Camp Competition (Excellence Prize) Awarded by CEO, Gyeongnam Creative Economy Innovation Center
02/2026	GNU RISE Shared Achievement Forum – Outstanding Startup Club (Commendation) Awarded by Director, GNU RISE Program
12/2025	Southeast Regional University Startup Competition (Grand Prize) Awarded by Director of Startup Support Center, Gyeongsang National University
11/2025	Data Venture Problem-Solving Challenge (Excellence Award) Awarded by Director of Big Data Innovation Convergence University, Seoul National University
11/2025	GNU Pioneer Leaders Startup Idea Hackathon (Grand Prize) Awarded by Director, GNU Startup Support Center
09/2025	Gyeongnam Regional University Startup Club Competition (Grand Prize) Awarded by Director, GNU Startup Support Center
07/2025	GNU-Startup Pioneer (Excellence Award) Awarded by Director, GNU Startup Education Innovation Center
06/2025	Convergence Data Utilization Startup Competition (Public Institution Head Award) Awarded by President of Korea SMEs and Startups Agency
05/2025	GNU Pioneer Leaders Startup Ideathon (Grand Prize) Awarded by Director, GNU Startup Support Center
01/2025	ICT Convergence Competition (Generative AI Track) (Grand Prize) Awarded by President of Gyeongsang National University
01/2025	Winter Big Data Tableau Competition (Encouragement Prize) Awarded by CEO, Planit Partners
12/2024	AWS GenAI Schumpeter Hackathon (Grand Prize) Awarded by CEO of AWS Korea
12/2024	AWS GenAI Schumpeter Rocket Pitch (Excellence Award) Awarded by CEO of AWS Korea
12/2024	Creative Convergence 3D Printing Idea Competition (Encouragement Prize) Awarded by Dean, GNU College of Engineering
09/2024	National University Development Program Student Project Competition (Excellence Award) Awarded by Director, GNU Dept. of Industrial & Systems Engineering

PROFESSIONAL CERTIFICATES AND TRAINING

2025	SQL Developer (SQLD) Korea Data Agency (KDATA), Seoul, South Korea
2025	Drone License (Level 4) Korea Data Certification Authority
2024	Certificate of Computer Literacy (Level 2) Human Resources Development Service of Korea